

Superstore Sales Analysis Project

From Messy Data to Dashboard

Basic Summary of Investigation, Decisions, and Actions

1) Data Cleaning Phase

Null checks

Scanned for null and blanks values using Power Query column profiling, which reveals nulls as 'Empty' (none found). Applied Trim and Clean transformations across fields to standardise formatting and remove excess whitespace and non-printable characters.

Data Type Corrections

Fields containing numeric data were in text format - changed to numeric type to allow accurate aggregation, calculation, and sorting. Quantity changed to whole number type, while financial fields - Sales, Discount, and Profit - changed to decimal number to give full precision.

Order Date and Ship Date were text fields in US date format. Converted them using the appropriate locale format, to prevent parsing errors resulting from regional date-format differences.

Errors

Verified that there were no data-quality issues, using Power Query's 'Keep Errors' functionality to expose any potential problems systematically, rather than visually scanning (impractical for a dataset of this size).

Handling Duplicates

Investigated potential duplicate transactions by identifying a uniqueness key (Order ID and Product ID) and grouping them accordingly. Verified suspected duplicates against full transaction details, identifying one confirmed duplicate (removed) and numerous legitimate repeated records (retained).

2) Feature Engineering Phase

Data Transformation – Splitting Customer Names

The Customer Name field contained two distinct pieces of data - the customer's first and last names. Split the column into two to store these fields separately, preserving multi-part last names.

Extracted Useful Date Component Parts

Order Date required in its existing (full) format, but some component parts were useful for analytics. Extracted the year, the month (as numeric), and the month name from the Order Date, to enable analysis based on this data.

Created Profit Margin Metric

Created a profit margin metric (calculated from $\text{Profit} \div \text{Sales}$) to measure profitability relative to revenue generated, enabling the identification of transactions by their profit status (profitable, break even, or loss-making).

Created Profit Status Classification

Created a Profit Status classification column using conditional logic based on profit margin values, categorising transactions as profit-making, break-even, or loss-making. Retained transactions with a negative profit, as they represented legitimate unprofitable transactions rather than data errors, and were subsequently analysed to identify patterns in loss-making sales.

3) Pivot Analysis Phase

Created several PivotTables to explore profitability from different perspectives.

Sales & Profit by Category compared revenue and profit over the three main product categories, enabling assessment of relative profitability.

Profit by Region analysed yearly profit performance over the four regions, highlighting regional trends over the four years.

Profit Impact by Profit Status summarised sales and profit values for profit-making, loss-making, and break-even transactions by year. It formed the basis for more focused visualisations separating sales and profit impact.

Transactions Count by Profit Status showed the number of profit-making, loss-making, and break-even transactions for each year, providing operational context alongside the financial analysis. Helped determine if loss-making transactions comprise rare, high-impact losses, or frequent, low-margin transactions.

4) Dashboard Development Phase

Created several charts to illustrate the findings.

Sales & Profit by Category showed differences in profitability efficiency over the three main product categories, revealing that high sales revenue did not always correspond to proportionally high profit.

Profit by Region highlighted regional differences in profitability over the 2014-2017 period, revealing the strongest-performing regions.

Sales by Profit Status illustrated the proportion of revenue generated by profitable, break-even, and loss-making transactions, highlighting that not all sales activity contributed positively to profitability.

Profit by Profit Status revealed that overall profitability increased over the period at a greater rate than loss-making activity, indicating improving financial performance as the business grew.

Transactions Count by Profit Status illustrated that most transactions remained profitable throughout the four years, even as total transaction volume increased.